

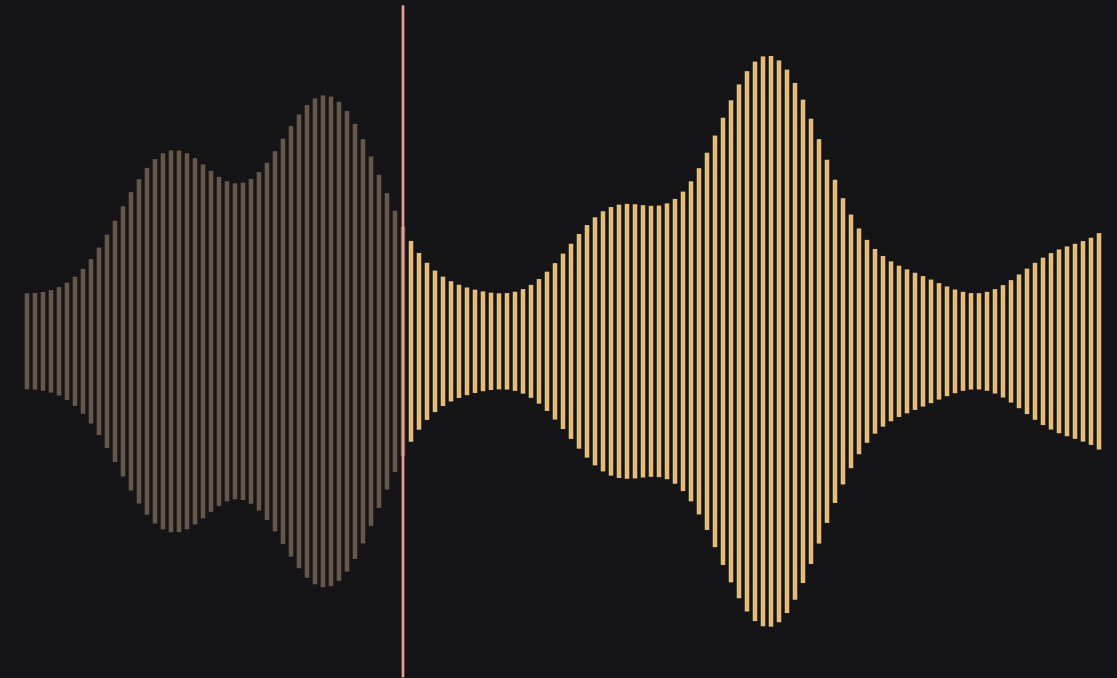
SonicCraft

Hear the change.
See the signal.

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Python desktop audio editor



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Why SonicCraft?

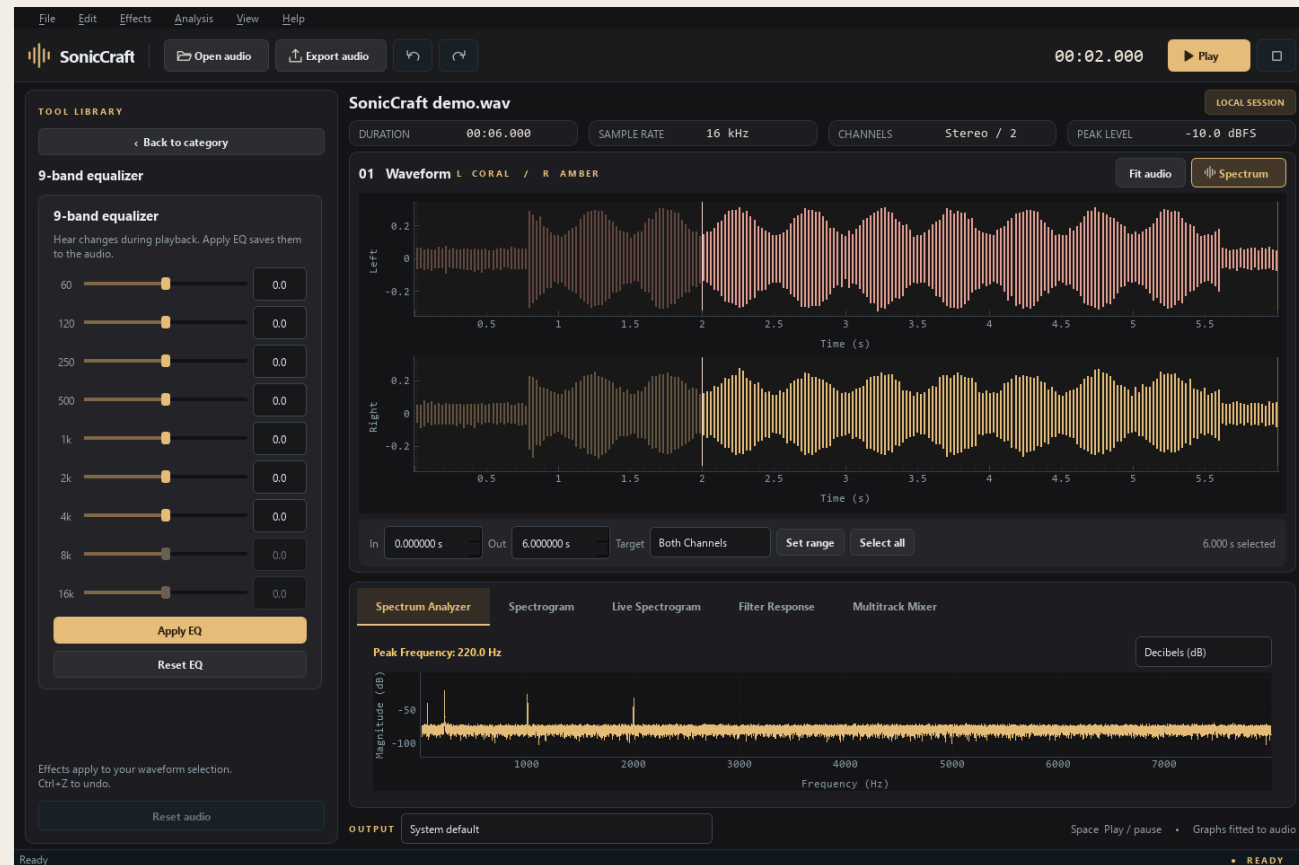
Make an audio change visible and audible.

THE PROBLEM

Editing sound by ear alone is difficult.

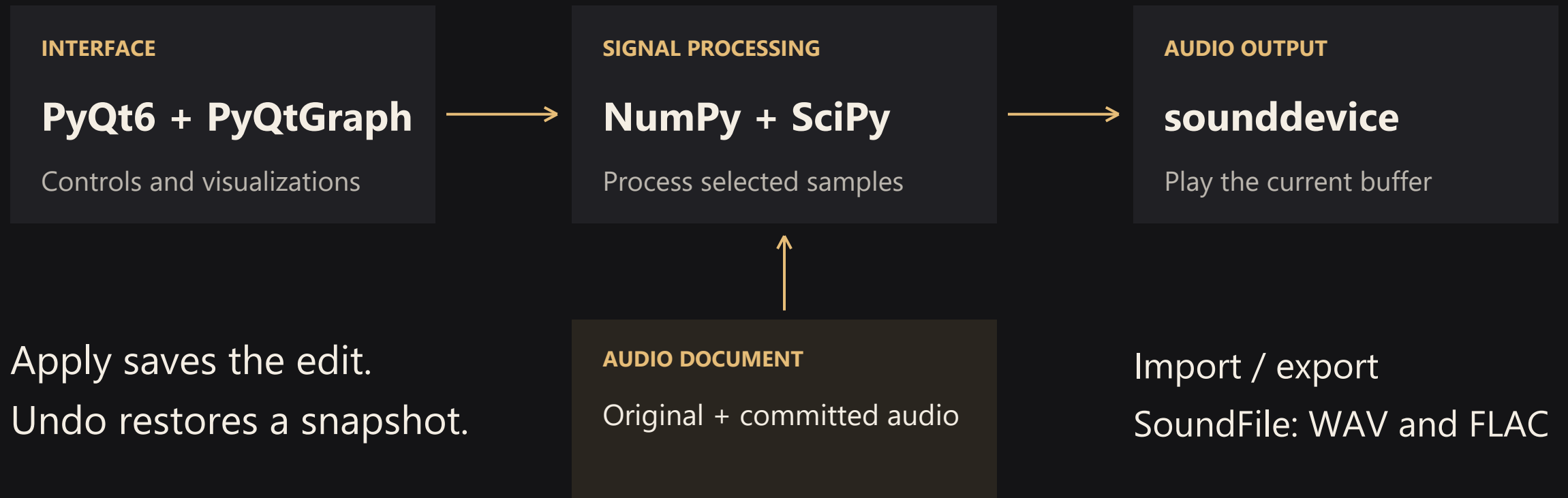
OUR APPROACH

Inspect the signal.
Change a control.
Hear the result.



How the application works

One audio document connects editing, processing and playback.

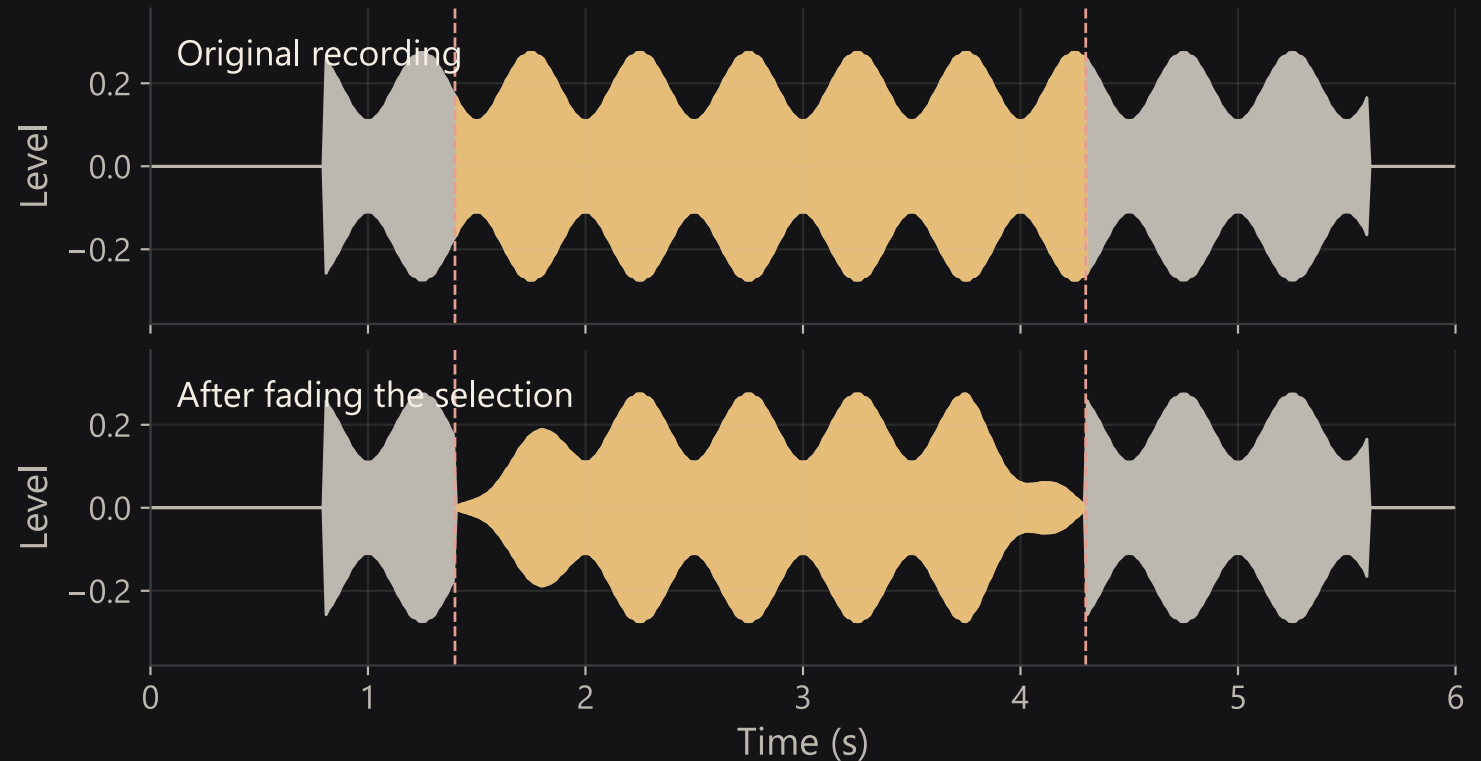


Edit & Dynamics

Shape the selected part of a recording.

EDIT & DYNAMICS

- Select, trim or remove a time range.
- Adjust level or normalize the peak.
- Fade edges for a smooth entrance or exit.



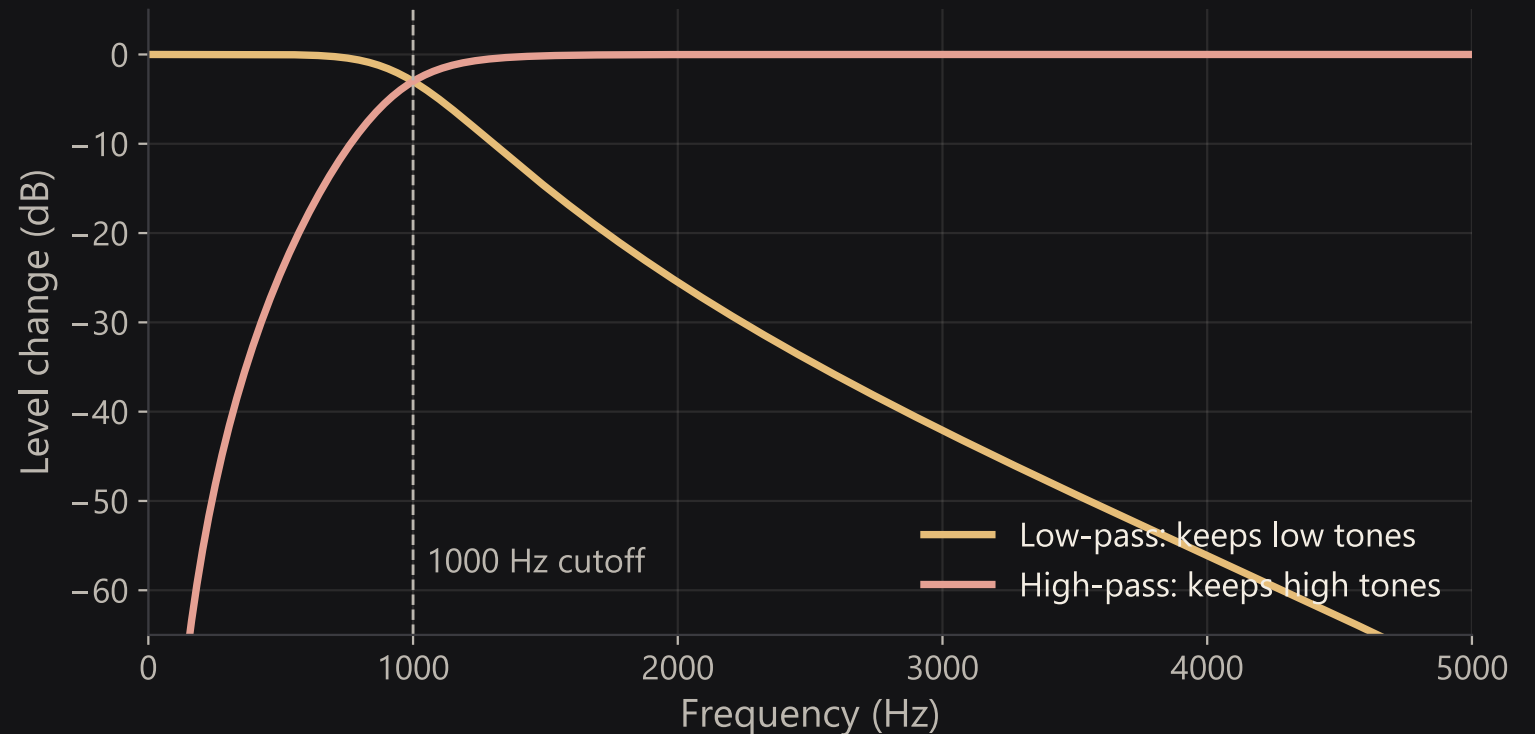
A real clip: the highlighted interval fades at both ends.

Frequency filter

Keep wanted frequencies and reduce others.

FILTERS & EQUALIZER

- Choose low-pass, high-pass or band filters.
- Set the cutoff to shape what passes.
- Hear the result while adjusting.



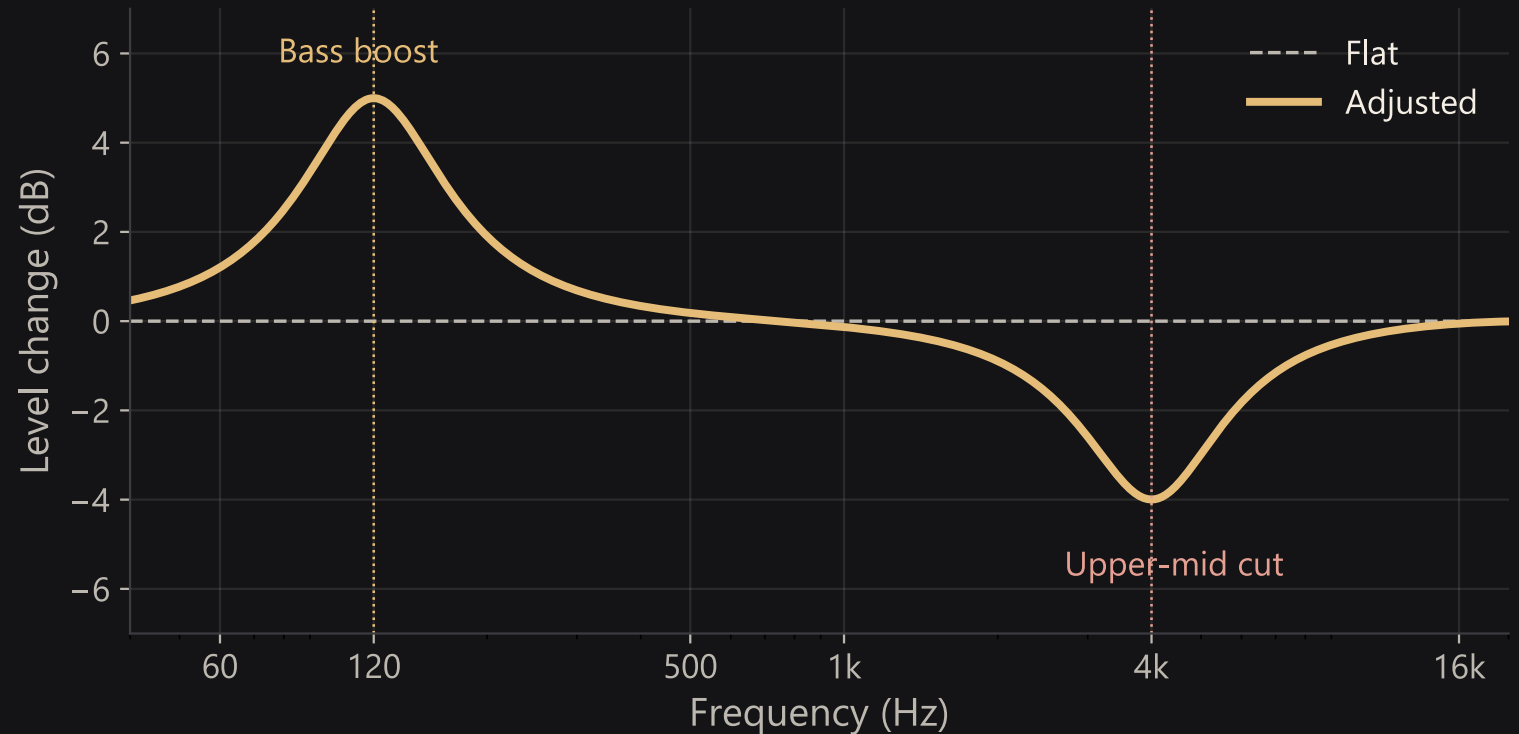
Two filter settings at the same cutoff frequency.

9-band equalizer

Change the tonal balance of the recording.

FILTERS & EQUALIZER

- Move a band to boost or reduce its range.
- Combine bands across bass, mids and treble.
- Apply EQ to save what you hear.



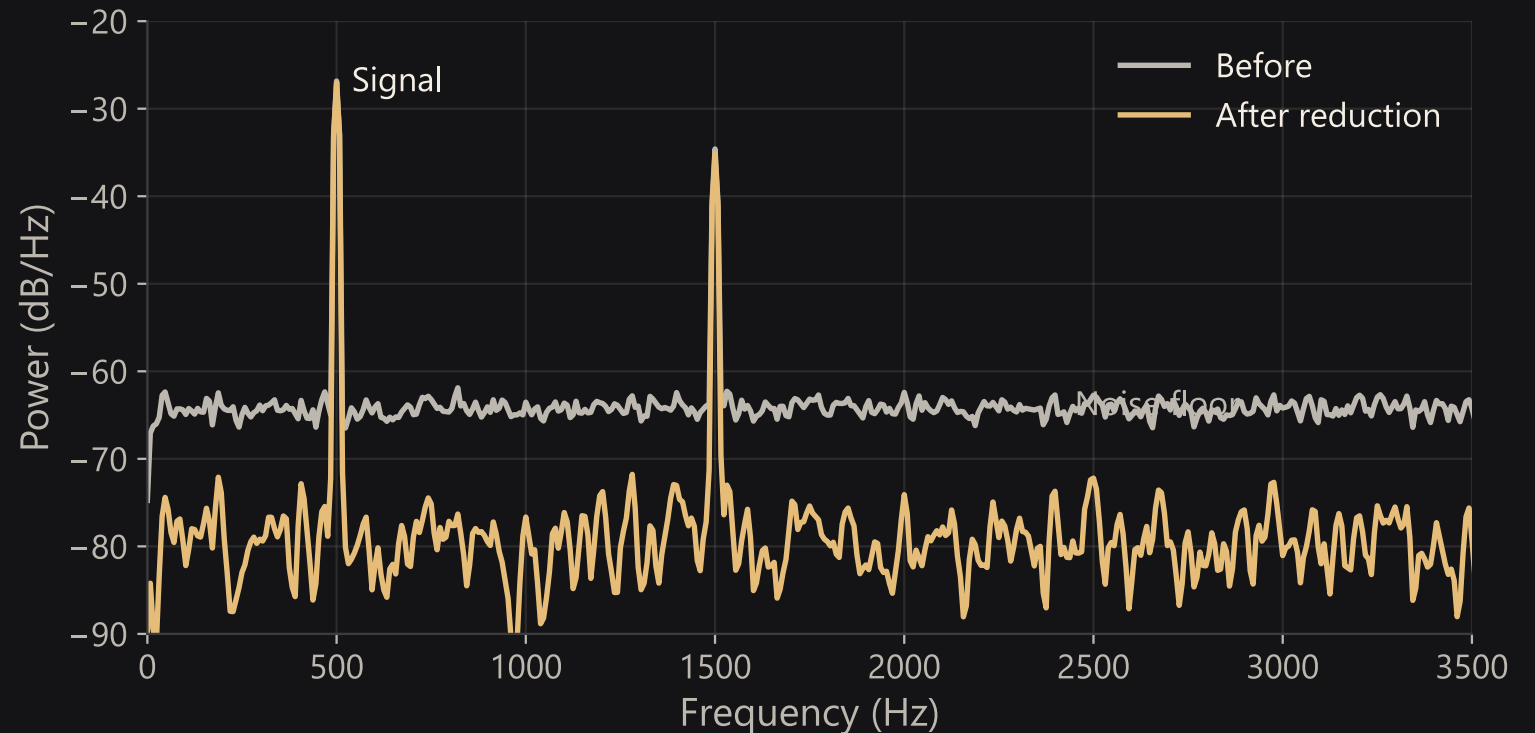
Example: bass boost and upper-mid cut; other bands stay flat.

Noise reduction

Reduce steady background sound.

NOISE REDUCTION

- Capture a section containing only noise.
- Estimate its frequency pattern.
- Reduce matching noise in the selection.



The noise floor drops while two signal peaks remain.

Spectrum analyzer

See which frequencies exist in the selection.

SPECTRUM ANALYZER

- Select audio and calculate its FFT.
- Read the strongest frequency peaks.
- Compare the spectrum before and after edits.



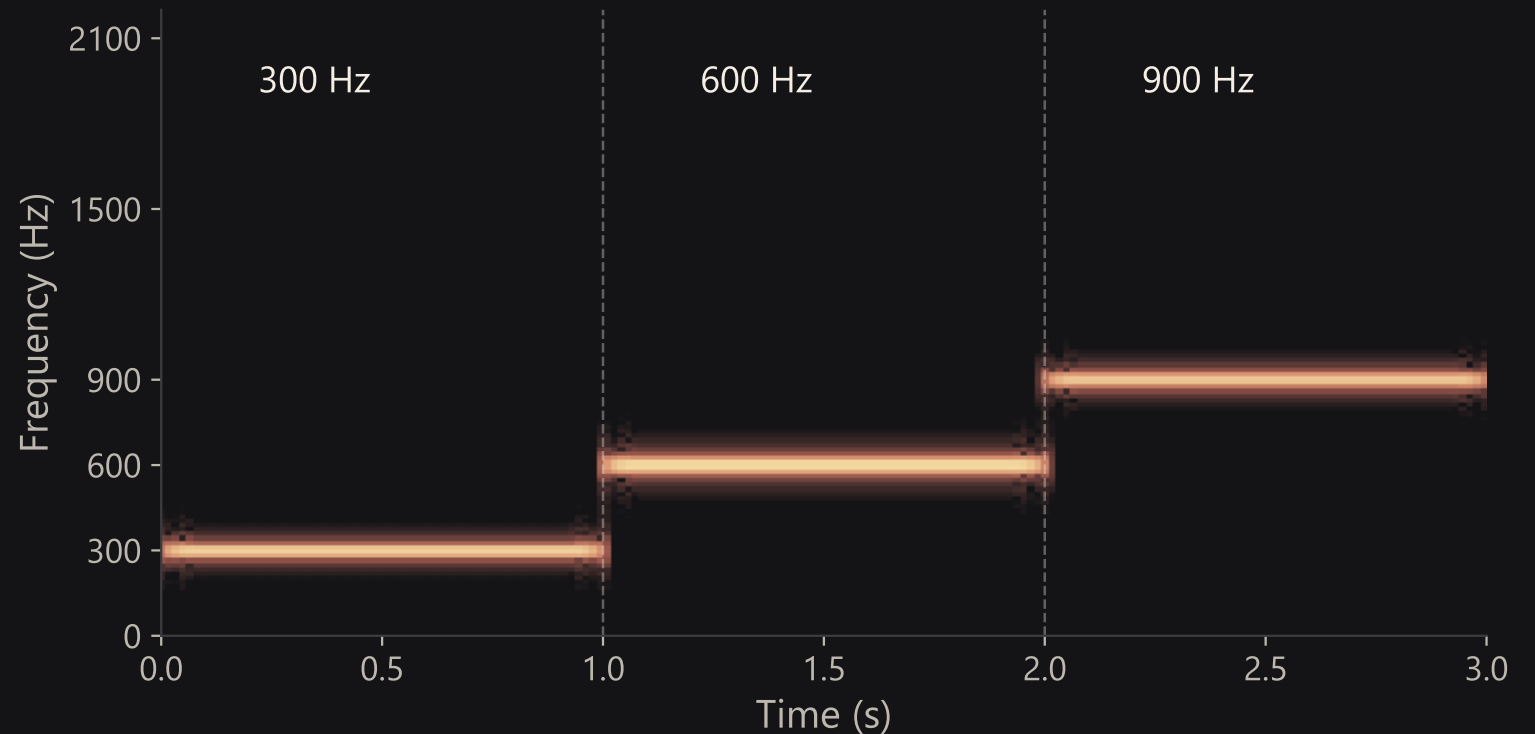
A two-tone example produces peaks at 500 Hz and 1500 Hz.

Spectrogram

See how frequency content changes with time.

SPECTROGRAM

- Time runs left to right; frequency runs upward.
- Bright bands show strong components.
- Find when a tone or sound occurs.



Three notes played in sequence: 300, 600 and 900 Hz.

Live spectrogram

Monitor microphone sound as it arrives.

LIVE SPECTROGRAM

- Capture microphone audio in short blocks.
- Refresh a rolling time-frequency view.
- Optionally record the input for editing.

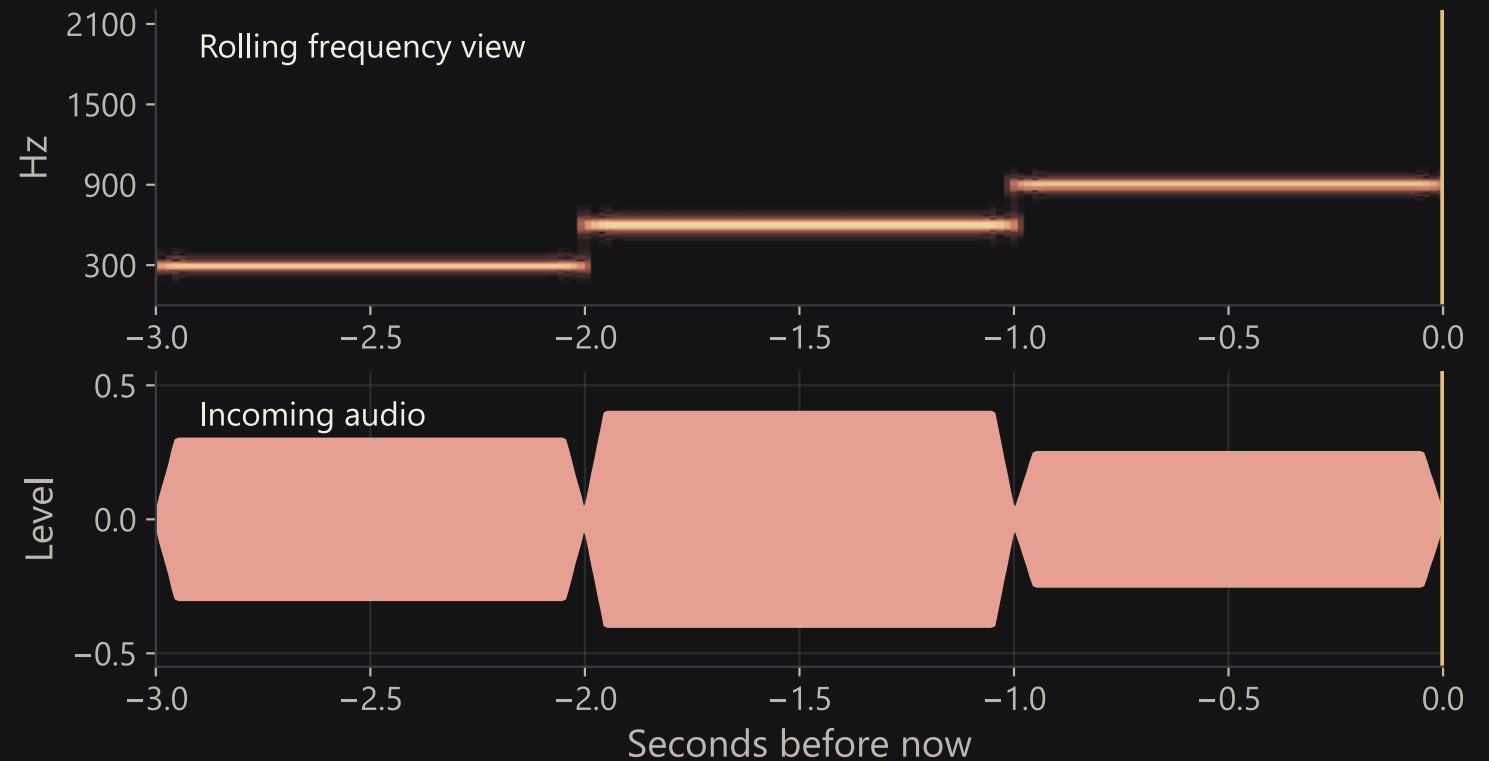
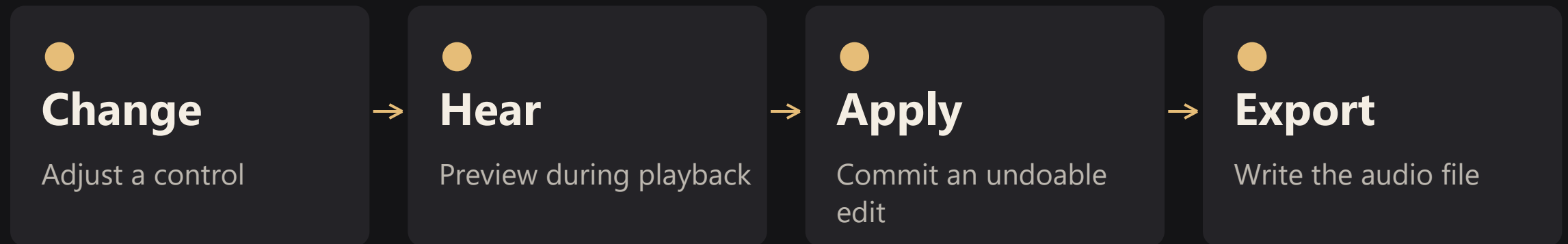


Illustration with generated notes; the live view uses microphone input.

Preview, apply and reset

A clear path from adjustment to saved audio.



TWO RESET SCOPES

Tool reset

Clears this tool's settings and preview.

Reset audio

Restores the original audio and all settings.

SONICCRAFT

Explore the project in code.

Source, documentation and a reproducible demo

github.com/mehrab4226/SonicCraft